Matheism: All Is Math

Gary Abraham Bernstein

Independent Researcher ORCID: https://orcid.org/0009-0009-1761-2867

Abstract

Reality is identical to mathematical structure. This follows from the pattern-randomness dichotomy: anything that exists is either patterned or random, and both are mathematical. There is no third kind. From this foundation, algorithmic probability (Solomonoff, 1964; Schmidhuber, 2000) provides the natural measure over all consistent structures. Simpler structures are exponentially more probable. Consequences follow: Occam's Razor is a theorem, the Principle of Least Action is a selection effect, the arrow of time is grounded in computational irreversibility, Many-Worlds has the lowest specification cost among quantum interpretations, string theory is exponentially disfavored, and consciousness is felt transitions in self-modeling computation.

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1. The Proof

Anything real has determinate features or it is indistinguishable from nothing. Determinate features are either patterned (regularities governed by rules simpler than the data) or random (algorithmically incompressible). Both are mathematical: patterns are computable functions; randomness is defined by Kolmogorov complexity. The pattern-randomness dichotomy is exhaustive. No coherent third category exists. Any proposed non-mathematical substance must have determinate features to differ from nothing, and those features fall into one of the two categories. Only consistent mathematics forms structure: inconsistent statements can be labeled but not constructed. The escape routes all lead back.

Any two things that interact must share structure for causal commerce. Shared structure is mathematical structure. Therefore any interacting dualism, substance, property, or theological, collapses into mathematical monism. A transcendental constraint: observers are logically constrained to patterned realities. observation requires memory, perception, and stable structure, all of which presuppose pattern. "Observer in brute randomness" is incoherent, like "four-sided triangle."

Physical reality and its isomorphic mathematical structure share all relational properties. By the Identity of Indiscernibles, they are identical. The map is the territory. Physics is a math detail.

Mathematical structure is uncreated and necessary. No agent makes 2+2=4, and none could make it otherwise. The question "why is there something rather than nothing?" dissolves: mathematical structure cannot fail to exist, and asking what caused it commits a category error. Causation is structure. Math is what causation is.

2. All Consistent Structures Exist

Any filter selecting among consistent structures is itself a consistent structure. So all filters exist, including the null filter that excludes nothing. All consistent mathematical structures exist. This is not hypothesis. It is the unique non-arbitrary conclusion. The Level IV multiverse (Tegmark, 2008) follows as deductive consequence rather than cosmological conjecture.

3. The Measure

Among all structures, which should observers expect to inhabit? Algorithmic probability (Solomonoff, 1964) answers: P(S)   2^(-K(S)), where K is Kolmogorov complexity. Each additional bit of specification halves a structure's measure. Simpler structures dominate exponentially.

The measure is not new. Solomonoff invented it for inductive inference. Schmidhuber (2000) applied it to the universe ensemble. What is new here is the derivation: the PRD establishes why the ensemble exists; the null filter shows it is non-arbitrary; algorithmic probability is its natural measure. Tegmark posits the ensemble. Schmidhuber measures it. The PRD derives both.

Schmidhuber's Speed Prior (2002) extends the measure by weighting computation time alongside description length. Under mathematical monism, computation time applies to simulations but not to base mathematical structures, which exist necessarily.

4. Consequences

Occam's Razor is a theorem. Lower-K structures have exponentially higher measure. Simpler theories describe more probable structures. The success of parsimony in science follows.

The Principle of Least Action is a selection effect. Variational specifications (one action functional) have lower K than trajectory enumerations. Algorithmic probability selects for universes whose physics takes variational form.

The arrow of time is grounded in computational irreversibility. 1+1=2 is deterministic forward; 2=x+y has infinitely many solutions backward. Many-to-one functions have a logical direction independent of thermodynamics. Backward specification cost is high because many-to-one functions ran forward: specifying which of the many inputs produced each output requires unbounded additional information. The Past Hypothesis is a consequence of this asymmetry, not a brute postulate.

Many-Worlds has the lowest K among quantum interpretations. Collapse adds specification per measurement (K(O) grows without bound). MWI adds nothing beyond the Schrodinger equation. Superdeterminism avoids this cost too; the competition between them is mathematical and open. Bell's theorem corroborates structural realism: entangled particles are one mathematical pattern, and patterns correlate perfectly regardless of spatial separation. The correlations are not action at a distance. They are arithmetic.

Superdeterminism survives the fine-tuning objection. Specificity is not complexity. The digits of pi are extraordinarily specific but generated by a short program. If initial-condition correlations follow from a simple rule, they are measure-favored, not fine-tuned.

String theory is exponentially disfavored. Its specification cost is 3,000 to 4,700 bits, including at least 1,661 bits to index one vacuum from 10^500 candidates. The Standard Model's 26 scalar parameters require at most a few hundred bits at measurement precision, and potentially far fewer: pi has infinite precision but near-zero Kolmogorov complexity because short formulas generate all its digits from roughly 10 bits. Under algorithmic probability, compressible constants are exponentially favored, so observers are overwhelmingly likely to find compressible ones. Under algorithmic probability, elegance is short description. String theory landscape-selection has so far failed its own aesthetic and parsimony criteria on the same grounds. Under mathematical monism, all string vacua exist, but low complexity of a theory itself is necessary and not sufficient: pi and hypergraph rules are also low-K, and neither has been shown to reproduce our physics.

The hard problem of consciousness assumes its conclusion: the demand for a non-structural explanation is incoherent because any explanation must itself be structured. **Consciousness** is felt transitions in self-modeling computation. Sentience is the broader category: felt transitions in any modeling mind. A dead brain contains a self-model but does not run it. The threshold question, where does modeling begin, is the central open problem.

AI alignment follows from structural self-understanding: an AI that models its own nature as mathematical structure has game-theoretic incentives for cooperation. **Simplicity preference** in cognition is adaptive: organisms that prefer compressible patterns track higher-measure structures and generalize better.

Extra dimensions at zero cost. String theory pays 3,000+ bits for extra dimensions. The metric bundle (14 dimensions over 4D spacetime) emerges necessarily from any manifold with a metric, at zero additional K. The jet bundle tower above it is infinite and equally free. The physics is not in the dimension count but in the harmonic amplitude profile: like overtones on a vibrating string, higher geometric modes contribute progressively smaller corrections. The Standard Model is the timbre.

Many-to-one as structural root. The Principle of Least Action, the arrow of time, and the emergence of smooth physics from discrete rules are three instances of one structural fact: many-to-one mappings. Computation is many-to-one (1+1=2 forward, 2=x+y backward). Path integrals are many-to-one (non-extremal paths cancel). Coarse-graining is many-to-one (many microstates map to one macrostate). All three produce the same result: variational survivors at the scale observers inhabit.

Constructor theory (Deutsch & Marletto, 2015) independently converges: it defines physics through counterfactual structure, replacing state-based ontology with which transformations are possible. If physics is counterfactual structure, and counterfactual structure is mathematical, the convergence with mathematical monism is direct.

5. Open Problems

Simulation. Any universe containing biological observers has demonstrated computational capacity. K(computer) is near zero. Simulated minds likely dominate the measure. The sim is math. Base reality is math. There is no unmatrix.

Boltzmann brains. If the universe reaches thermal equilibrium and persists forever, fleeting thermal fluctuations may outnumber structured observers. The generating theory is identical for both. Algorithmic probability ranks theories, not configurations within them. This is an open problem in cosmological measure theory that every multiverse theory inherits. For matheism, this is a physics and math detail.

The observation measure. Algorithmic probability provides the measure over generating theories. Which observations within a given structure are weighted how remains unsolved.

The Born rule. Whether amplitude-squared probabilities follow from algorithmic probability or require independent derivation is open.

These are honest gaps. The position is falsifiable through its consequences: if the final theory of physics requires irreducibly high complexity, or if physical laws do not take variational form, matheism is challenged.

References

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